Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 6. Estimates of reduction in risk of severe malaria

Nicholas J. Savill Philip J. Spence

1 Introduction

Severe malaria episode data from Gonçalves et al. (2014) demonstrates that the risk of a clinical episode becoming severe decays exponentially with the number of blood infections:

$$\Pr(\text{clinical episode becomes severe on } n^{\text{th}} \text{ blood infection}) = \rho_{\text{severe}} e^{-\delta_{\text{severe}}[n-1]}$$

where ρ_{severe} is the risk of a clinical episode becoming severe on the first infection and δ_{severe} is the decay in the risk with the number of blood infections.

Very similar estimates of the decay rate and the risk on first infection have been obtained by Muñoz Sandoval et al. (2025), and by ourselves. In Notebook S4 we estimated ρ_{severe} to be $4.2\% \pm 0.5\%$ and δ_{severe} to be 0.12 ± 0.03 — which corresponds to a halving of risk every 5.7 blood infections.

2 Consequences of this model for severe malaria and mortality age patterns

In Figure 1 we plot severe malaria and mortality age patterns for 1 and 5 blood infections per year using the simple model described above.

At 5 blood infections per year (red lines) severe malaria episode rate and mortality rate drop rapidly in the first few years of life resulting in roughly 25% of children having at least one severe episode and about 1.7% of children dying.

At 1 infection per year (purple lines) the severe episode rate and the mortality rate remain high into the teens and there is a second peak in deaths due to rising death rate with age after 4 years old (Reyburn et al. (2005); Notebook S7). Cumulative severe episodes rise more slowly and plateau at the same level as for 5 blood infections per year. Cumulative deaths also rise more slowly but reach higher levels than 5 blood infections per year, again due to the rising death rate with age.

So, although this simple model of exposure-only reduction in risk of severe malaria fits data from Gonçalves et al. (2014), it is not entirely consistent with other mortality age-pattern datasets as we show below.

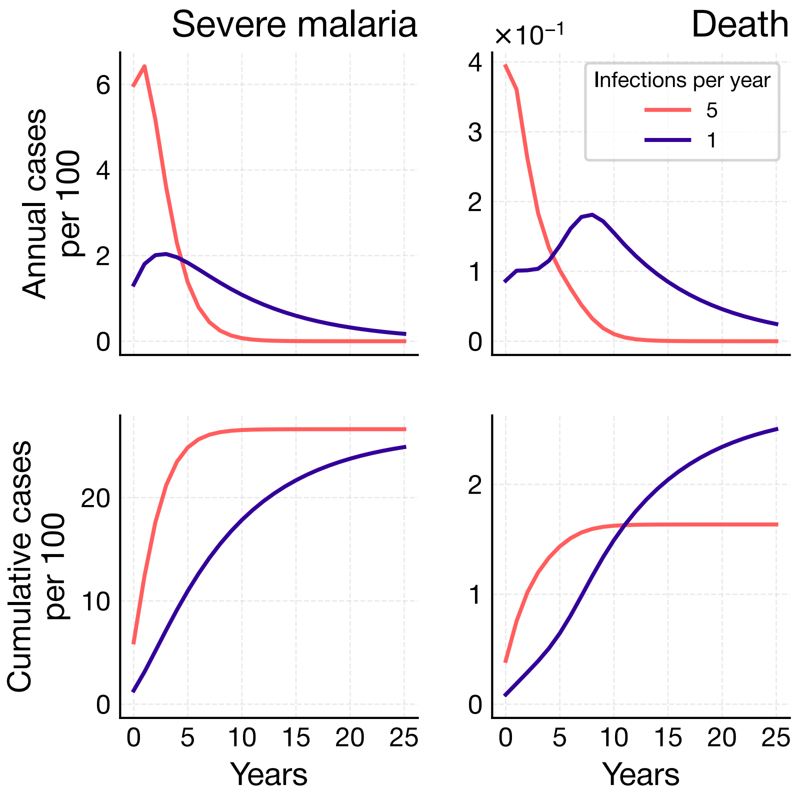

Figure 1: Severe malaria and mortality rates and cumulative episodes for 1 and 5 blood infections per year assuming exposure-only dependent reduction in risk of severe malaria.

3 Evidence in support of a more complex model of severe malaria immunity

Abdullah et al. (2007) published mortality age patterns based on verbal autopsies in seven sites with different transmission intensities. The open circles in Figure 2 show malaria-specific deaths per 1,000 person years at risk from 0-2 years of age (left-hand panels) and 2-14 years of age (right-hand panels). Kisumu and Manhiça have the lowest transmission intensities (actual rates were not given in Abdullah et al. (2007)). There is no indication of a substantial or prolonged second peak in any site and most deaths occur within the first four years of life.

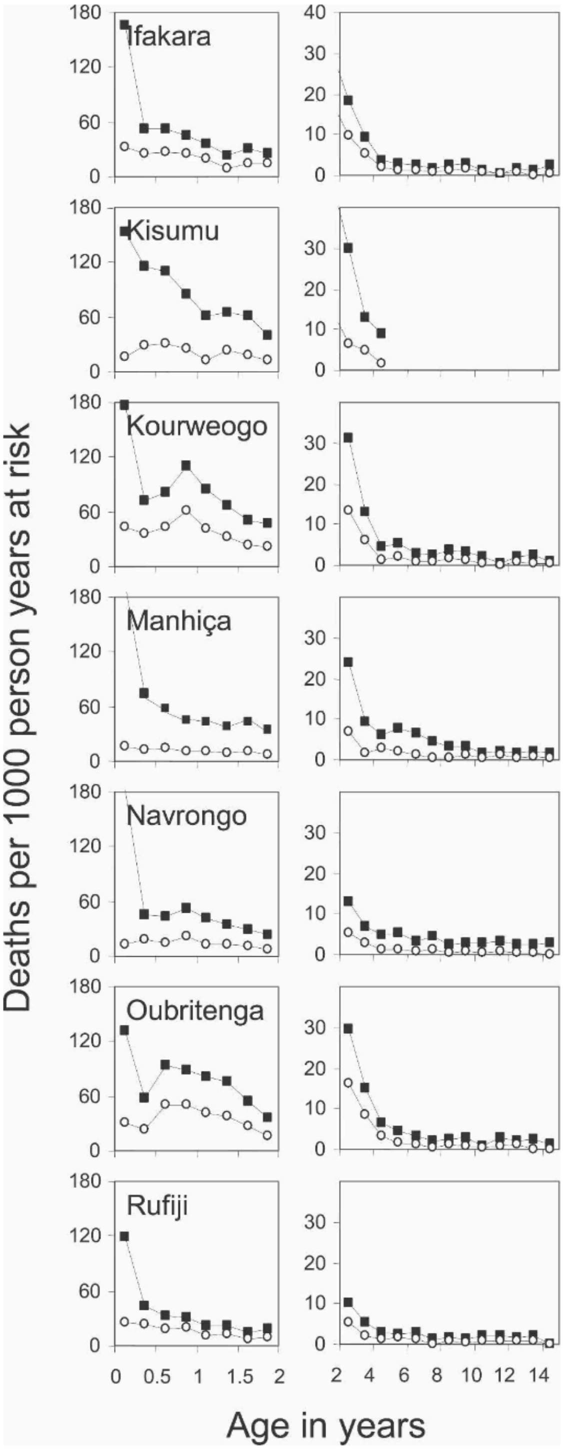

Figure 2: Age-specific mortality rates from Abdullah et al. (2007). Left panels: 0-2 year olds; right panels 2-14 year olds. Closed squares: all cause mortality; open circles: malaria-specific mortality. Permission to reproduce pending.

Cumulative probabilities of malaria-specific deaths by age for each site are shown in Figure 3. There are two features in this data not captured by our model. First, cumulative probabilities plateau at different levels for the different transmission intensities: Manhiça showing the lower cumulative probability. Second, the shoulder in the curves occurs at around three years of age irrespective of transmission intensity.

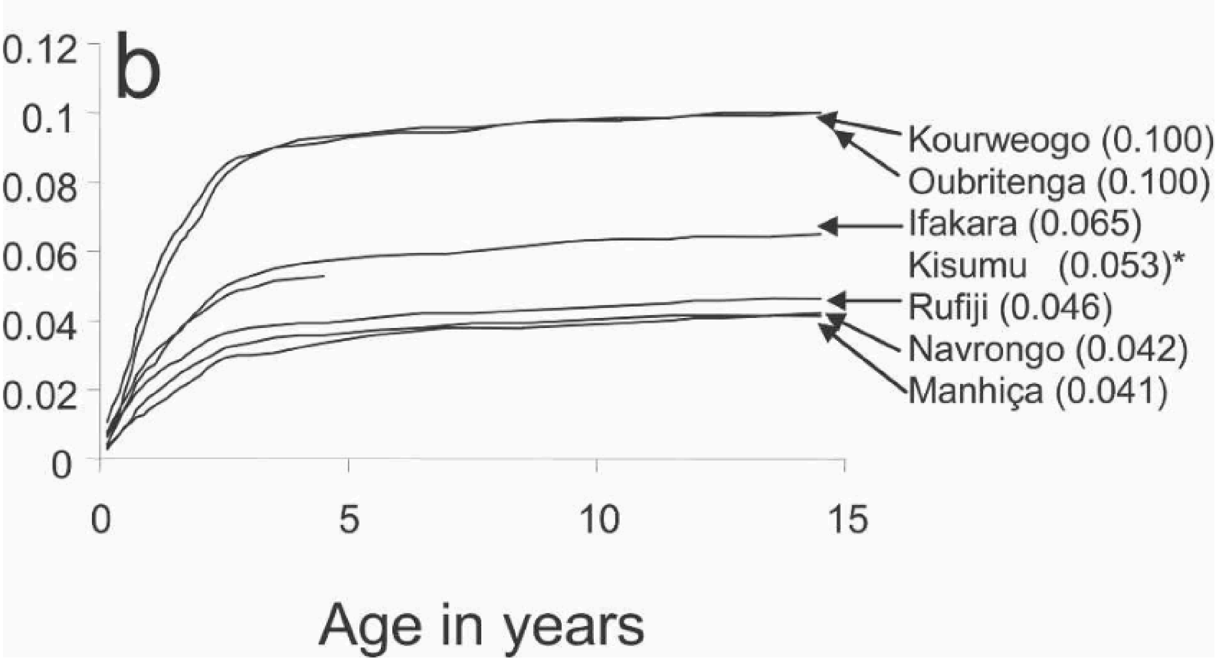

Figure 3: Cumulative probability of malaria specific mortality by given age. Figures in parenthesis are the overall probability of dying by age 15 years. For Kisumu the data analysed refer only to ages 0–5 years, and the figures in parenthesis are the probability of dying by age 5 years. Permission to reproduce pending.

Reyburn et al. (2005) recorded 1989 severe malaria episodes from hospital admissions from northeastern Tanzania in villages experiencing different transmission intensities, stratified as low, medium and high. Denominator data is not available so severe malaria rates cannot be calculated. However, cumulative age patterns are plotted in Figure 4. As for deaths, the shoulders in the curves occur in the under fives; a pattern not reproduced by our model.

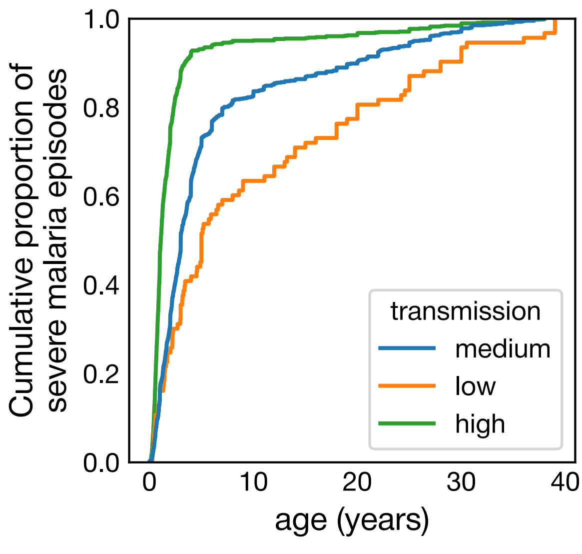

Figure 4: Cumulative age patterns of severe malaria from Reyburn et al. (2005).

4 A modified model with risk reducing with age-dependent exposure

A simple addition to make our model consistent with the observed severe malaria and death age patterns is to allow the rate of decay of risk of severe malaria to also be dependent on age. A possible modified model is

$$\Pr(\text{clinical episode becomes severe on } n^{\text{th}} \text{ blood infection at age } a) = \rho_{\text{severe}} e^{-[\delta_{\text{severe}}+\epsilon_{\text{severe}}a][n-1]}$$

where the new parameter ϵ_{severe} introduces age dependent reduction in risk with exposure.

Figure 5 demonstrates the effect ϵ_{severe} has on severe malaria and death age patterns when $\lambda = 1$ blood infection per year. In particular it suppresses the second peak in deaths in the five to ten year olds. It also reduces the cumulative number of severe malaria episodes and deaths. Although we have no data to formally fit ϵ_{severe}, based on the verbal autopsy data in Abdullah et al. (2007), a value of 0.05 per blood infection per year seems reasonable.

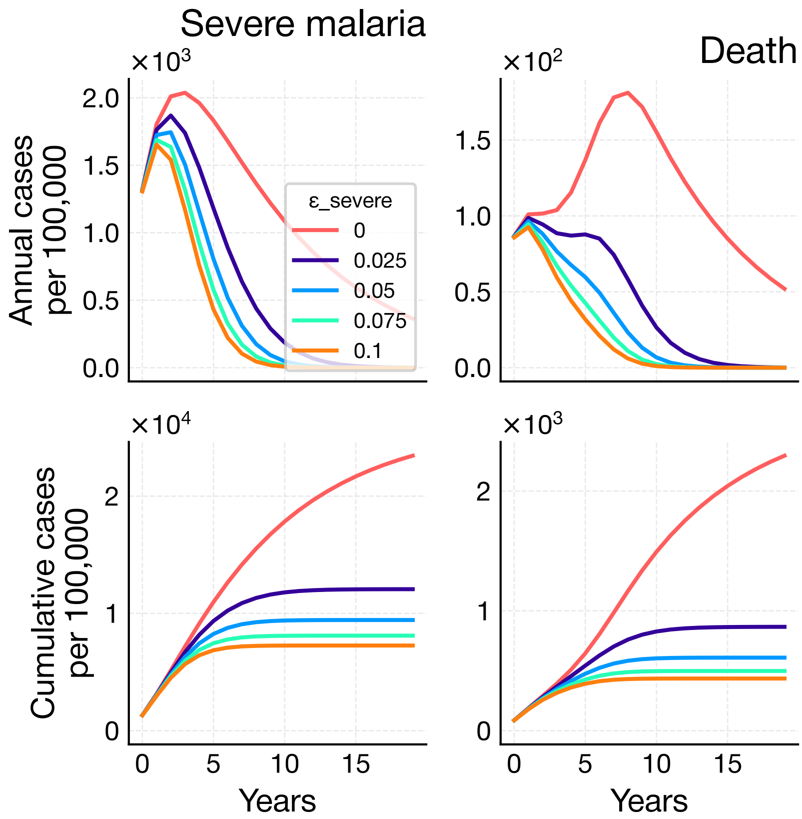

Figure 5: Effect of ϵ_{severe} on severe malaria and mortality rates for 1 blood infection per year.

Setting $\epsilon_{\text{severe}} = 0.05$ per blood infection per year, and using our estimated values of $\rho_{\text{severe}} = 0.042$ and $\delta_{\text{severe}} = 0.12$ per blood infection, we get risk curves as shown in Figure 6. Irrespective of age, the risk of a clinical episode on first blood infection becoming severe is 4.2%. However, older children with the same number of blood infections as younger children are less at risk of severe malaria.

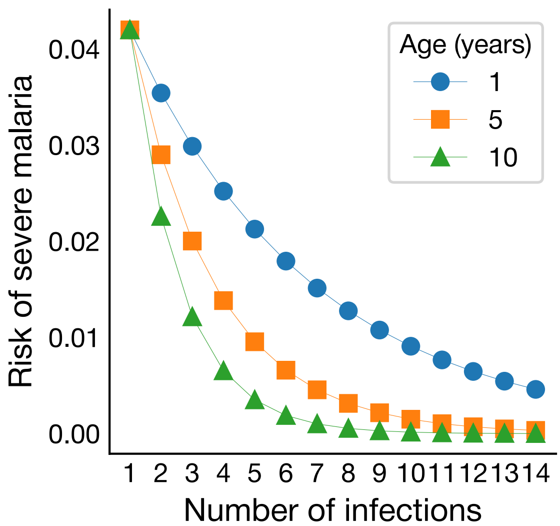

Figure 6: Risk of a clinical episode becoming severe depends on age and exposure.

For this modified model, the consequences for age patterns are shown in Figure 7. Now, compared to 5 blood infections per year, 1 blood infection per year:

1. does not exhibit a second peak in deaths
2. has lower cumulative severe malaria episodes and deaths
3. a similar age when cumulative severe malaria episodes and deaths plateau

There are, of course, several caveats with this model. First, we have no immunological evidence that risk decays with age. Second, we have no high quality epidemiological data across sites with different transmission intensities with which to estimate ϵ_{severe}.

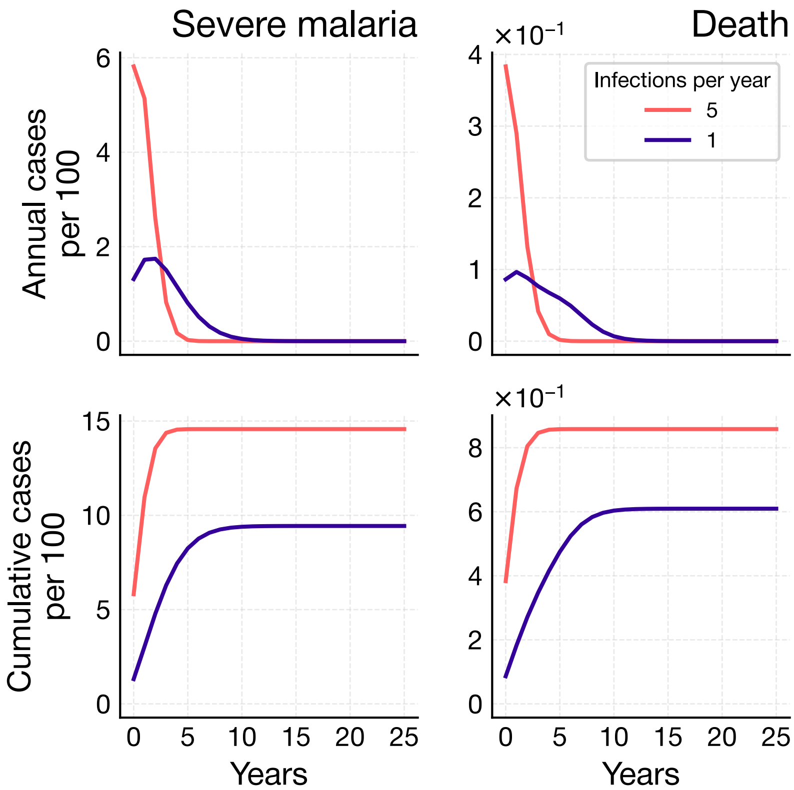

Figure 7: Severe malaria and mortality rates and cumulative episodes for 1 and 5 infections per child per year assuming age-dependent exposure reduction in risk of severe malaria.

5 Validation of severe malaria age patterns

Typical age patterns of severe malaria are shown in Figure 8.

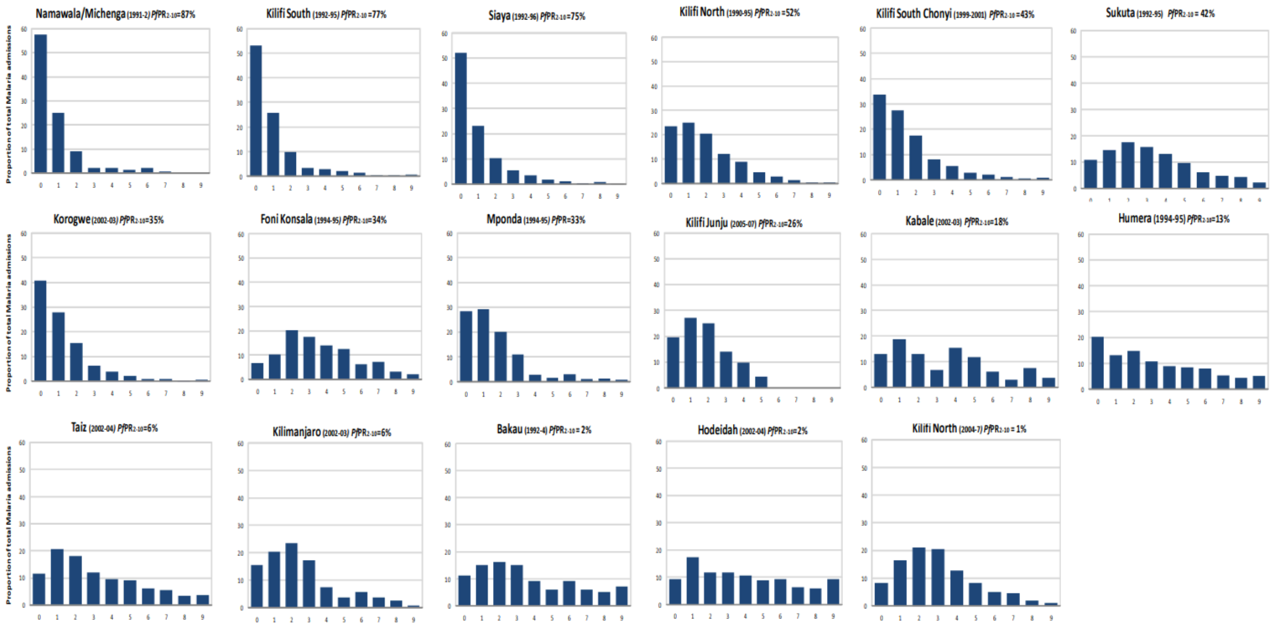

Figure 8: Age distribution of hospitalized malaria from 17 communities arranged by decreasing *Pf*PR$_{2\text{-}10}$ (Okiro et al. 2009). Permission to reproduce pending.

Simulations of our model with $\epsilon_{\text{severe}} = 0.05$ per blood infection per year for different blood infection rates are shown in Figure 9 are in good agreement with the observed age patterns of Okiro et al. (2009).

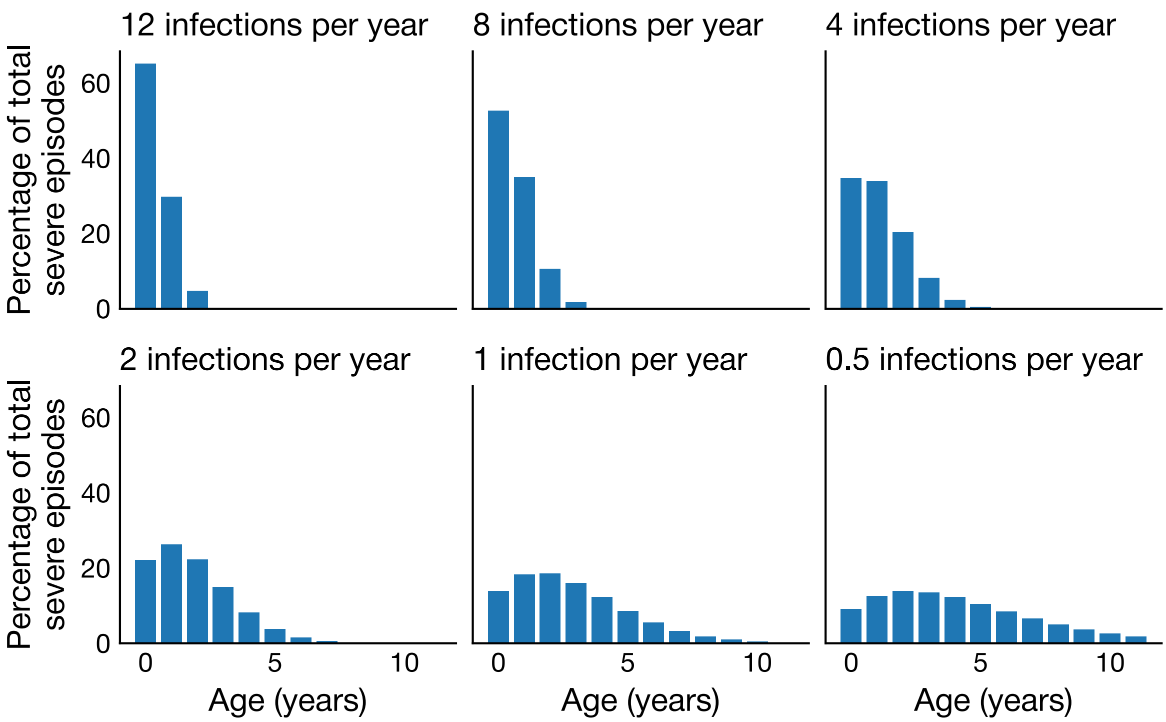

Figure 9: Model simulations of severe malaria age patterns for a range of transmission intensities using default parameters.

Bibliography

Abdullah, Salim, Kubaje Adazu, Honorati Masanja, Diadier Diallo, Abraham Hodgson, Edith Ilboudo-Sanogo, Ariel Nhacolo, et al. 2007. "Patterns of Age-Specific Mortality in Children in Endemic Areas of Sub-Saharan Africa". *American Journal of Tropical Medicine and Hygiene* 77 (6_Suppl): 99–105. https://doi.org/10.4269/ajtmh.77.6.suppl.99.

Gonçalves, Bronner P., Chiung-Yu Huang, Robert Morrison, Sarah Holte, Edward Kabyemela, D. Rebecca Prevots, Michal Fried, and Patrick E. Duffy. 2014. "Parasite Burden and Severity of Malaria in Tanzanian Children". *New England Journal of Medicine* 370 (19): 1799–1808. https://doi.org/10.1056/nejmoa1303944.

Muñoz Sandoval, Diana, Florian A. Bach, Alasdair Ivens, Adam C. Harding, Natasha L. Smith, Michalina Mazurczyk, Yrene Themistocleous, et al. 2025. "\textit{Plasmodium Falciparum} Infection Induces T Cell Tolerance That Is Associated with Decreased Disease Severity Upon Re-Infection". *Journal of Experimental Medicine* 222 (7): e20241667. https://doi.org/10.1084/jem.20241667.

Okiro, Emelda A., Abdullah Al-Taiar, Hugh Reyburn, Richard Idro, James A. Berkley, and Robert W. Snow. 2009. "Age Patterns of Severe Paediatric Malaria and Their Relationship to Plasmodium Falciparum Transmission Intensity". *Malaria Journal* 8 (1): 0–11. https://doi.org/10.1186/1475-2875-8-4.

Reyburn, Hugh, Redempta Mbatia, Chris Drakeley, Jane Bruce, Ilona Carneiro, Raimos Olomi, Jonathan Cox, et al. 2005. "Association of Transmission Intensity and Age with Clinical Manifestations and Case Fatality of Severe \textit{Plasmodium Falciparum} Malaria". *JAMA* 293 (12): 1461–70. https://doi.org/10.1001/jama.293.12.1461.